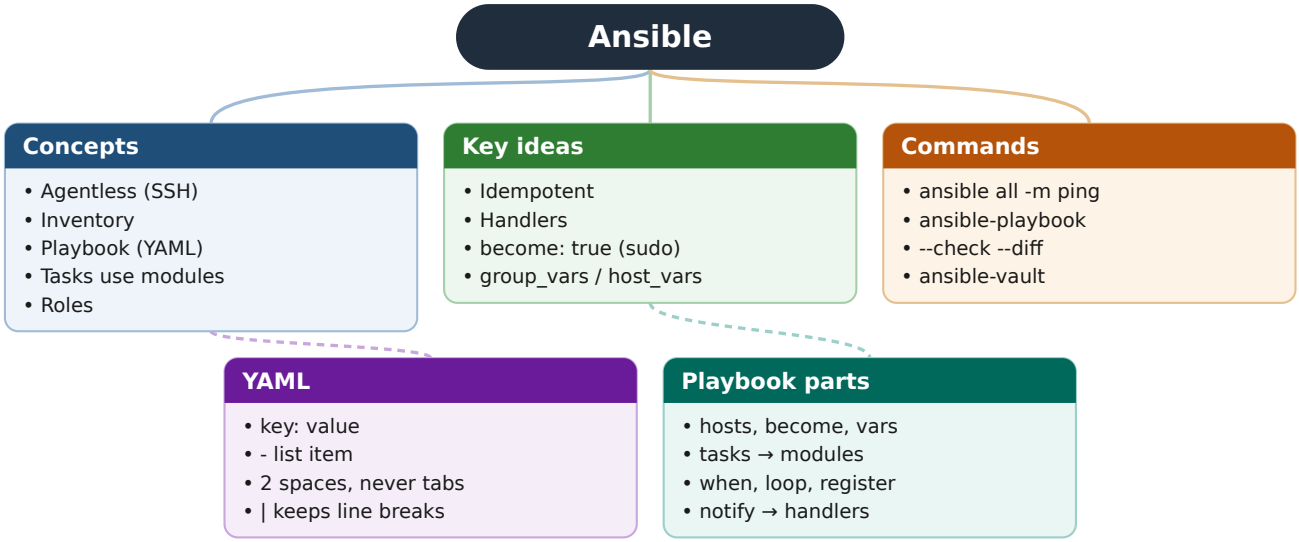
# 10. Ansible (Configuration Management)



**In this section:** 10.1 YAML: what is what · 10.2 Playbook anatomy: what is what




**Configuration management** = keeping servers configured the same way (packages, files, users, services) automatically. **Terraform builds the servers; Ansible configures what's inside them.**

| Term | Meaning |
|---|---|
| Agentless | Connects over SSH; nothing to install on target servers |
| Inventory | List of servers and groups |
| Playbook | YAML file with tasks to run on hosts |
| Task / module | One action, e.g. `apt`, `copy`, `template`, `service`, `user` |
| Role | A reusable, structured bundle of tasks, templates and variables |
| Idempotent | Running it again changes nothing if the server is already correct |
| Handler | A task that runs only when notified (e.g. restart nginx after config changes) |
| Ansible Vault | Encrypts secrets in Ansible files |

```
# inventory.ini
[web]
10.0.1.10
10.0.1.11

[web:vars]
ansible_user=ubuntu
```

```
# site.yml
- name: Configure web servers
  hosts: web
  become: true                    # use sudo
  tasks:
    - name: Install nginx
      ansible.builtin.apt:
        name: nginx
        state: present
        update_cache: true
    - name: Deploy config
      ansible.builtin.template:
        src: nginx.conf.j2
        dest: /etc/nginx/nginx.conf
      notify: Restart nginx
    - name: Ensure nginx is running
      ansible.builtin.service:
        name: nginx
        state: started
        enabled: true
  handlers:
    - name: Restart nginx
      ansible.builtin.service:
        name: nginx
        state: restarted
```

| Command | What it does |
|---|---|
| `ansible all -i inventory.ini -m ping` | Test the connection to all hosts |
| `ansible-playbook -i inventory.ini site.yml` | Run a playbook |
| `ansible-playbook ... --check --diff` | Dry run, show what would change |
| `ansible-vault encrypt secrets.yml` | Encrypt a secrets file |

| More Ansible | Purpose |
|---|---|
| `ansible-galaxy collection install amazon.aws` | Install community/vendor modules |
| `ansible web -m setup` | Show facts (OS, IPs) gathered from hosts |
| `group_vars/`, `host_vars/` | Variables per group / per host |
| `--limit web1`, `--tags nginx` | Run on one host / only tagged tasks |
| Dynamic inventory | Build the host list from AWS automatically (by tags) |

## 10.1 YAML: what is what

YAML is used by Ansible, Kubernetes, Docker Compose, GitHub Actions and Helm, so learn it once.

```
# Comment
name: web-server          # key: value   (a space after the colon is required)
port: 8080                # number
enabled: true             # boolean (true/false)
version: "1.10"           # quote values that look like numbers but are text
empty: null               # no value

servers:                  # LIST: each item starts with "- "
  - web1
  - web2

database:                 # MAP (dictionary): nested keys, indented
  host: db.internal
  port: 5432

users:                    # list of maps
  - name: alice
    groups: [admin, dev]  # inline list
  - name: bob
    groups: [dev]

script: |                 # | keeps line breaks (multi-line text, scripts)
  echo "line 1"
  echo "line 2"
description: >            # > folds lines into one long line
  This is a long
  sentence.
```

**Rules:** indentation = structure, **spaces only (never tabs)**, usually 2 spaces per level, items in the same list must line up. Check a file with `yamllint file.yml` or `python3 -c "import yaml,sys; yaml.safe_load(open(sys.argv[1]))" file.yml`.

## 10.2 Playbook anatomy: what is what

```
---                                  # optional start-of-document marker
- name: Configure web servers        # PLAY: a name for humans
  hosts: web                         # which inventory group to run on
  become: true                       # use sudo for tasks
  vars:                              # VARIABLES for this play
    app_port: 8080
    packages:
      - nginx
      - curl

  tasks:                             # TASKS run in order, top to bottom
    - name: Install packages         # always name tasks: shown in the output
      ansible.builtin.apt:           # MODULE (fully qualified name)
        name: "{{ packages }}"       # {{ }} = Jinja2 variable (quote it)
        state: present               # desired state, not a command → idempotent
        update_cache: true

    - name: Deploy nginx site config
      ansible.builtin.template:      # renders a Jinja2 template file with variables
        src: templates/site.conf.j2
        dest: /etc/nginx/sites-available/default
        mode: "0644"
      notify: Reload nginx           # trigger the HANDLER only if this task changed something

    - name: Check if the app answers
      ansible.builtin.uri:
        url: "http://localhost:{{ app_port }}/health"
        status_code: 200
      register: health               # save the result in a variable
      ignore_errors: true

    - name: Show the result
      ansible.builtin.debug:
        msg: "Health status: {{ health.status | default('unknown') }}"   # | = Jinja2 filter

    - name: Create users
      ansible.builtin.user:
        name: "{{ item }}"
        groups: www-data
        append: true
      loop:                          # LOOP: run the task once per item
        - alice
        - bob

    - name: Only on Ubuntu
      ansible.builtin.debug:
        msg: "This is {{ ansible_facts['distribution'] }}"   # FACTS gathered from the host
      when: ansible_facts['distribution'] == "Ubuntu"         # CONDITION

  handlers:                          # HANDLERS run once at the end, only when notified
    - name: Reload nginx
      ansible.builtin.service:
        name: nginx
        state: reloaded
```

**Template file** `templates/site.conf.j2` (Jinja2 = text with `{{ variables }}` and `{% logic %}`):

```
server {
    listen 80;
    location / {
        proxy_pass http://127.0.0.1:{{ app_port }};
    }
}
```

| Element | Meaning |
|---|---|
| Play | Maps a group of hosts to a list of tasks |
| Task | One step, using one module |
| Module | The tool that does the work ( `apt` , `copy` , `template` , `service` , `user` , `file` , `uri` , `command` ) |
| `state` | Desired result: `present` / `absent` , `started` / `stopped` / `restarted` |
| `{{ var }}` | Insert a variable (Jinja2) |
| `register` | Save a task's result |
| `when` | Run the task only if the condition is true |
| `loop` / `item` | Repeat a task for each item |
| `notify` / `handlers` | Run a follow-up (e.g. restart) only when something changed |
| `become` | Run with sudo |
| Facts ( `ansible_facts` ) | Information collected from each host (OS, IPs, memory) |

**Role layout** (reusable unit, created with `ansible-galaxy init nginx` ):

```
roles/nginx/
├── tasks/main.yml        # the tasks
├── handlers/main.yml     # handlers
├── templates/            # .j2 files
├── defaults/main.yml     # default variables (easy to override)
└── vars/main.yml         # fixed variables
```

Use it in a playbook with `roles: [nginx]` . **Avoid `command` / `shell` modules** when a real module exists: real modules are idempotent and report "changed" correctly.

**REMEMBER**

- **Terraform builds the house; Ansible furnishes it.**
- Idempotent = running it twice changes nothing the second time.